Calibration using FEM modeling:

1. Select N_p points $\mathbf{p}_i$ (3×1) for $i = 1,2,\cdots,N_p$ in the work space.
2. Perform N_I Finite Element Method (FEM) simulations using the input current vectors $\mathbf{I}_j = [I_{j1}\ I_{j2}\ I_{j3}\ I_{j4}\ I_{j5}\ I_{j6}]^T$ for $j = 1,2,\cdots,N_I$, ensuring that the combined current matrix $\mathbf{F} = [\mathbf{I}_1\ \ \mathbf{I}_2\ \ \cdots\ \ \mathbf{I}_{N_I}]$ has a rank of six. For each simulation.
3. Record the magnetic flux density $\mathbf{b}_{ij} = \mathbf{b}^{FEM}(\mathbf{p}_i, \mathbf{I}_j)$ (3×1) for $i = 1,2,\cdots,N_p$.
4. Record the sensor voltage v_{jk} for $k = 1,2,\cdots,6$ and for $j = 1,2,\cdots,N_I$.
5. Introduce a characteristic length scale $\hat{\ell}$, relating the geometry of the hexapole tweezers to the discrete locations of the six magnetic charges in the lumped-parameter model.
6. Define a nondimensionalized 3×6 spatial function matrix $\mathbf{S}(\overline{\mathbf{p}}_i; \hat{\ell})$ for each point. Perform spatial normalization:

$$\overline{\mathbf{p}}_i = \mathbf{p}_i/\hat{\ell} \text{ for } i = 1,2,\cdots,N_p.$$

$$\overline{\mathbf{p}}_c = [\overline{\mathbf{p}}_{c1}\ \ \overline{\mathbf{p}}_{c2}\ \ \overline{\mathbf{p}}_{c3}\ \ \overline{\mathbf{p}}_{c4}\ \ \overline{\mathbf{p}}_{c5}\ \ \overline{\mathbf{p}}_{c6}] = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\mathbf{S}(\overline{\mathbf{p}}_i; \hat{\ell}) = \left[\underbrace{\frac{\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_1}}{\|\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_1}\|^3}}_{\mathbf{s}_{i1}} \quad \underbrace{\frac{\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_2}}{\|\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_2}\|^3}}_{\mathbf{s}_{i2}} \quad \cdots \quad \underbrace{\frac{\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_6}}{\|\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_6}\|^3}}_{\mathbf{s}_{i6}} \right] (3 \times 6) \text{ for } i = 1,2,\cdots,N_p.$$

7. Cost function:

$$J(\hat{\ell}) = \sum_{j=1}^{N_I} \left\{ \sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{b}_{ij} - \sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{S}(\overline{\mathbf{p}}_i; \hat{\ell}) \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\overline{\mathbf{p}}_i; \hat{\ell}) \mathbf{S}(\overline{\mathbf{p}}_i; \hat{\ell}) \right)^{-1} \sum_{i=1}^{N_p} \mathbf{S}^T(\overline{\mathbf{p}}_i; \hat{\ell}) \mathbf{b}_{ij} \right\}$$

8. $Minimize\ J(\hat{\ell}) \Rightarrow \hat{\ell}$ (Initial guess: $\hat{\ell} = \ell_{design}$)
9. Calculate $\mathbf{g}_j = \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\overline{\mathbf{p}}_i; \hat{\ell}) \mathbf{S}(\overline{\mathbf{p}}_i; \hat{\ell})\right)^{-1} \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\overline{\mathbf{p}}_i; \hat{\ell}) \mathbf{b}_{ij}\right)$ (6×1) for $j = 1,2,\cdots,N_I$.
10. Define a $6 \times N_I$ matrix:
    $\mathbf{G} = [\mathbf{g}_1\ \ \mathbf{g}_2\ \ \cdots\ \ \mathbf{g}_{N_I}]$
11. Determine $\hat{g}_B \overline{\mathbf{K}}_I$: $\hat{g}_B \overline{\mathbf{K}}_I = \mathbf{H} = \mathbf{G}\mathbf{F}^T(\mathbf{F}\mathbf{F}^T)^{-1}$
    $\hat{g}_B = \frac{6}{5} h_{11}$ and $\overline{\mathbf{K}}_I = \frac{5}{6 \cdot g_{11}} \mathbf{H}$

$F = \phi R_a = NIR_a$

Hall-sensor-based hexapole model: $d_k v_{jk} = \frac{k_m}{\ell^2} \hat{q}_{jk}$ for $k = 1,2,\cdots,6$ and for $j = 1,2,\cdots,N_I$

$$\mathbf{b}_{ij} = \sum_{k=1}^{6} d_k v_{jk} \underbrace{\left(\frac{\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{ck}}{\|\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{ck}\|^3} \right)}_{\mathbf{s}_{ik}}$$

Model parameter estimation:

$$\boldsymbol{\varepsilon}_{ij}(\mathbf{d}) = \mathbf{b}_{ij} - \sum_{k=1}^{6} d_k v_{jk} \mathbf{s}_{ik} = \mathbf{b}_{ij} - \mathbf{S}_i \mathbf{V}_j \mathbf{d},$$

where $\mathbf{S}_i = [\mathbf{s}_{i1} \quad \mathbf{s}_{i2} \quad \cdots \quad \mathbf{s}_{i6}]$, $\mathbf{V}_j = \mathrm{diag}(v_{j1} \quad v_{j2} \quad \cdots \quad v_{j6})$, and $\mathbf{d} = [d_1 \quad d_2 \quad \cdots \quad d_6]^T$.

$$\mathbf{d} = \left(\sum_{i=1}^{N_p} \sum_{j=1}^{N_I} \mathbf{V}_j \mathbf{S}_i^T \mathbf{S}_i \mathbf{V}_j \right)^{-1} \left(\sum_{i=1}^{N_p} \sum_{j=1}^{N_I} \mathbf{V}_j \mathbf{S}_i^T \mathbf{b}_{ij} \right)$$

$$\mathbf{d} = \left(\sum_{j=1}^{N_I} \mathbf{V}_j \left(\sum_{i=1}^{N_p} \mathbf{S}_i^T \mathbf{S}_i \right) \mathbf{V}_j \right)^{-1} \left(\sum_{j=1}^{N_I} \mathbf{V}_j \left(\sum_{i=1}^{N_p} \mathbf{S}_i^T \mathbf{b}_{ij} \right) \right)$$

Six-parameter model:

The three axes in the actuator coordinate system: $[\hat{\mathbf{u}} \quad \hat{\mathbf{v}} \quad \hat{\mathbf{w}}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

The direction of the magntic pole direction: $\hat{\mathbf{a}}_k$

Define a nondimensionalized 3×6 spatial function matrix $\mathbf{S}(\overline{\mathbf{p}}_i; \ell)$ for each point, wherein ℓ is the design value. Perform spatial normalization:

$\overline{\mathbf{p}}_i = \mathbf{p}_i/\ell$ for $i = 1,2,\cdots, N_p$.
$\overline{\mathbf{p}}_c = [\mathbf{p}_{c1}/\ell \quad \mathbf{p}_{c2}/\ell \quad \mathbf{p}_{c3}/\ell \quad \mathbf{p}_{c4}/\ell \quad \mathbf{p}_{c5}/\ell \quad \mathbf{p}_{c6}/\ell]$
$\overline{\mathbf{p}}_c = [\hat{\mathbf{u}} + e_1\hat{\mathbf{a}}_1 \quad -\hat{\mathbf{u}} + e_2\hat{\mathbf{a}}_2 \quad \hat{\mathbf{v}} + e_3\hat{\mathbf{a}}_3 \quad -\hat{\mathbf{v}} + e_4\hat{\mathbf{a}}_4 \quad \hat{\mathbf{w}} + e_5\hat{\mathbf{a}}_5 \quad -\hat{\mathbf{w}} + e_6\hat{\mathbf{a}}_6]$
$\mathbf{e} = [e_1 \quad e_2 \quad e_3 \quad e_4 \quad e_5 \quad e_6]$

$$\mathbf{S}(\overline{\mathbf{p}}_i; \mathbf{e}) = \left[\underbrace{\frac{\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_1}}{\|\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_1}\|^3}}_{\mathbf{s}_{i1}} \quad \underbrace{\frac{\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_2}}{\|\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_2}\|^3}}_{\mathbf{s}_{i2}} \quad \cdots \quad \underbrace{\frac{\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_6}}{\|\overline{\mathbf{p}}_i - \overline{\mathbf{p}}_{c_6}\|^3}}_{\mathbf{s}_{i6}} \right] (3 \times 6) \text{ for } i = 1,2,\cdots, N_p.$$

18-parameter model:

$\overline{\mathbf{p}}_c = [\hat{\mathbf{u}} + \mathbf{e}_1 \quad -\hat{\mathbf{u}} + \mathbf{e}_2 \quad \hat{\mathbf{v}} + \mathbf{e}_3 \quad -\hat{\mathbf{v}} + \mathbf{e}_4 \quad \hat{\mathbf{w}} + \mathbf{e}_5 \quad -\hat{\mathbf{w}} + \mathbf{e}_6]$